5 Ways Books Can Empower Play: A Psychophysiological Design Study

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ABSTRACT
We gave 60 children (ages 6 to 9) a book designed to encourage play with a new toy. Using eye-tracking glasses, psychophysiological sensors, and ethnographic methods, we co-created with young readers, exploring how a book could be designed to encourage them to play. Our results showed that a will to play should not always be assumed – many children did not play with their toys despite the book directly prompting them to. However, children engaged with play when books directly referenced the child's toy, included parents, or included a digital camera that prompted them to tell a story.

Keywords
Literacy, Play, Human Centered Design, Psychophysiology

BACKGROUND
In the science fiction novel *The Diamond Age*, Neal Stephenson describes an electronic book with the power to teach and inspire play. Through the book's stories and prompts, Nell, a poor girl, practices somersaults to train self-defense and learns to read through playing games. Is it possible to make books like Nell's that empower children to learn through play?

Research suggests that play can be a strong way to learn, from cognitive processing and self-regulation (Vygotsky & Cole, 1978) to social and emotional development (Elkind, 2007) to story comprehension and narrative competence (Roskos et al., 2010). However, with a strong push toward core academic skills, schools are rapidly reducing time for play (Hirsh-Pasek et al., 2009; E. Miller & Almon, 2009).

Furthermore, play interventions tend to be focused on children ages 3 to 5, leaving older children out (Bubikova-Moan et al., 2019; Roberts, 1980). A review of play and literacy (Roskos et al., 2010) found only one study looking at the literacy benefits of play for children in first grade and above. The elementary classrooms past kindergarten we work in do not have allotted play centers or playtime.

Research suggests that apps or books that encourage learning through play should have five core traits; they should be joyful, active, meaningful, social, and iterative (Hirsh-Pasek et al., 2015; Zosh et al., 2018). Specifically, within socio-dramatic play, Smilansky (1968) suggests children should be taught how to play with modeling, verbal guidance, thematic fantasy training, and imaginative play training. Using these guidelines, is it possible for a book to empower children to learn through play?

METHODOLOGY:
Over 10 months, we worked with 60 children ages 6 to 9 in after-school programs and in their homes in Denver, Colorado. We used a co-creation and iterative design (Cobb et al., 2003) to fully understand what young readers need to be encouraged to play.

Children were gifted a new toy of their choice, sometimes days in advance, other times during the research session. At the research session, the children were given a book about their new toy. We asked the child to read the book and then left the room, observing the session via eye tracking. We were careful never to ask children to play with the toy. If children did not read the book within five minutes of us leaving, we asked the child to read again or asked the parent to read with the child.

Figure 1. Matilda wears eye-tracking glasses and reads a prototype book.

When willing, children wore video-recording eye-tracking glasses during each session to help reveal how the story was read and whether the children played unprompted and unobserved (Tobii Technology AB, Sweden). The eye-tracking cameras allowed the researchers to be in a separate room and observe remotely, reducing the influence an observing scientist would have on the play session; our previous research showed that children are highly influenced to play "correctly" with an observer watching them (Hedman, 2008). At other times, a camera was set up on a stand to record the session.

When willing, some children wore skin-conductance sensors (Hedman, 2008) to track their emotional responses during reading and play, as we have done in other studies (Hedman, 2010, 2018, 2011). Skin conductance is a marker of sympathetic arousal and can help show the level of a child's emotional or cognitive engagement (Boucsein, 2011).

Additionally, we conducted open-ended interviews after each intervention. Children were asked to describe how they used the books and toys. Based on our remote observations, we would prompt children to tell us about certain behaviors: "Can you tell us what happened on this page?" We would then ask children about their preferences for different designs and ask what they would add to the books to better encourage learning through play.

Description of Intervention

In each research session, we gave children a toy (plastic dinosaurs, a fidget cube, Play-Doh, a slime-making kit, a LEGO Set, or Pokémon action figures). We also gave children a book that referenced their toy. Each week, based off children's feedback, we modified the book to better encourage learning through play. Over the course of the study, 24 different books were developed and tested, all of which can be reviewed here: www.buildempathy.com/empowerplay. Books were 18 to 64 pages long, typically with one or two sentences per page. Most pages included pictures of the toy, and comic bubbles were used extensively. Stories directly referenced the toys children were given and prompted children to play with the toys. Text was written at a second-grade reading level. Parents or researchers read with children unable to read the text independently.

In our later interventions, children were given the book five minutes after the toy to better understand how children naturally play. We ended recording when the child told the researcher they were done. This allowed us to better observe children's natural play after reading as well.

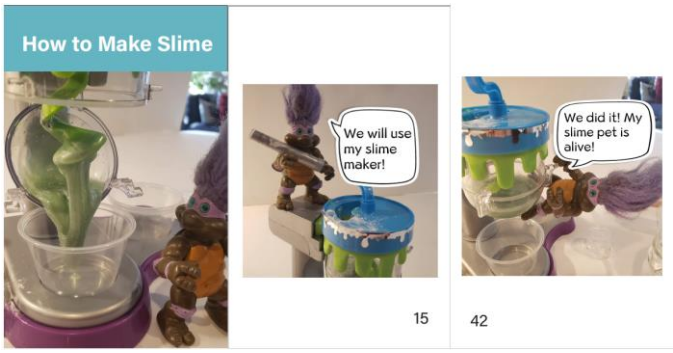

Figure 2. An example of a prototype book, this one about making slime.

Definition of Play

We designed our books to encourage socio-dramatic play, where children role-play their own imaginative story (Fleer, 2016; Smilansky, 1968). Children were identified as playing when they interacted with their toys and verbally described a story that took place. Sometimes, children mechanically interacted with toys – spinning wheels or moving a sword up and down without telling a story. These mechanical interactions were not considered play in our study.

RESULTS:

Socio-dramatic play often does not occur when children receive a new toy and book.

When I visited Daniel, he was so excited to participate in our study that he wore a suit. I forgot a prototype book at the previous house, so I gave Daniel a LEGO model to play with until I returned.

Thirty minutes later, upon returning, the LEGO model was built and sitting on a shelf. Daniel was with his brother watching television. I asked Daniel what he and his brother did after they built the model. He showed me the model chopping with a sword. "We had a fight," he said. They only played for a few seconds.

After our session with Daniel, I asked participants to sit with their toy for five minutes before and after reading to see how often unprompted play occurred. Typically, little to no play occurred before or after the session. Occasionally, children would spin the toy's wheels or weapons, but storytelling, for the most part, did not naturally happen during our research. Following are two more representative stories of what typically happened in our interventions.

Brady loved Pokémon and personally requested a story about the action figures. He looked at the small plastic figures of Pikachu and Bulbasaur and then set them aside, not telling any story with the characters, even when the book prompted Brady to tell a story.

Figure 3. Brady reads a prototype book without playing with the Pokémon toys.

After reading, Matilda moved her LEGO surfboard up and down to pretend it was going on the waves, but that was the extent of the play. Matilda's book prompted her to finish the story: "Can you help Layla and Emma throw a surprise birthday party?" When Matilda's mother urged her to play out that prompt, Matilda shook her head vigorously and turned the page herself.

The transition between reading and play can prevent play.

When children told stories with their toys, the play was unstructured and child-driven, whereas practicing reading is often assigned and observed by a teacher and requires hard work. Often, our books prompted children to interact and play with the toys, but the children did not transition to play. The young readers may not have interpreted the directions correctly or did not want to switch from reading to play. Switching tasks can be difficult for this age group (Cepeda et al., 2001).

Other times, children struggled transitioning from play to reading. When Emilio was given a bucket of dinosaurs, he made a dinosaur dance party but did not want to open the book, even when directed by the teacher.

Figure 4. The dinosaur book is buried under the toys and was never opened, even after prompting.

Similarly, when Bea and Layla were playing with Play-Doh, they refused to open the book that told a story about making a castle, even when directed by the researcher. Children often treated reading as a separate activity from play.

Our sensors showed emotional differences between reading and play as well. Kyle's emotional response to reading (Figure 5) dramatically differs to his emotional response to play (Figure 6), where he is acting out a story on his own about a LEGO mech attacking a town.

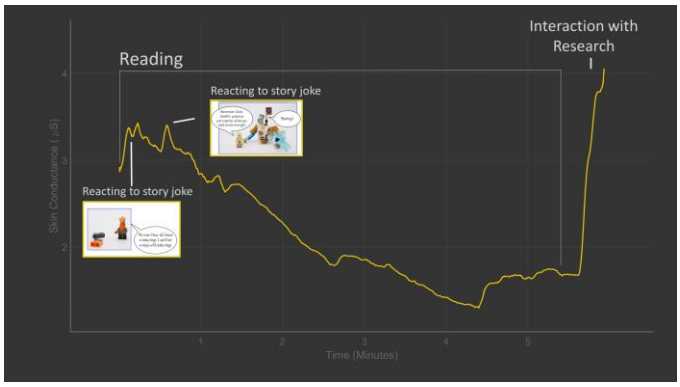

Figure 5. Kyle's skin conductance during reading (low arousal).

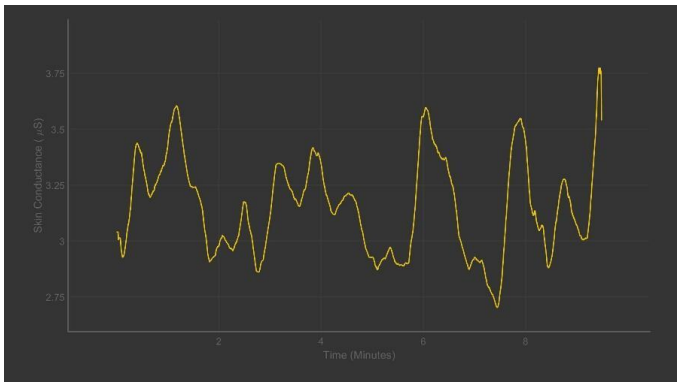

Figure 6. Kyle's skin conductance during play (high arousal).

While he reads, Kyle's skin conductance is low with few responses, and his body in a calm state. Our previous research showed similar types of responses to reading in general (Hedman, 2021). However, when Kyle is playing on his own, he is constantly emotionally responding – the play is active and emotionally engaging.

As discussed in the next section, two of our most effective interventions directly address this transition challenge. Parents were often the catalysts to transitioning children from reading to play, guiding the child in switching activities. Our digital camera intervention transitioned reading and play by making reading more playful, with interactive questions and a requirement for children to take a picture to continue reading.

FIVE DESIGN RECOMMENDATIONS FOR BOOKS TO ENCOURAGE PLAY

Before our research intervention, Coulomb's dad mentioned that his son had never told a story with the LEGO sets he received; the toys were only built and put on a shelf. After reading a prompt from one of our prototype books, Coulomb acted out a story of a LEGO mech kicking out its driver and having the treasure chest drive the mech down the dangerous mountain. The following week, when Coulomb bought a new LEGO Star Wars set, without researchers or prompting, he told another story with his new spaceship. Below we list five ways our prototype books successfully encouraged children like Coulomb to play.

Figure 7. Coulomb plays out a story of a mech marching a treasure chest down a mountain.

A. Provide a background story for toys (Meaning)

Smilansky (1968) suggests teaching children background information about their play theme can enhance play. Children naturally tell the stories of Batman and Cinderella because they know their stories: they know the character names, how the characters behave, and the stories that revolve around those characters (Paley, 2005).

To provide children with background knowledge, our books included profiles for each character, detailing information about the toys to help inspire play.

Figure 8. Descriptive text directly gives character hooks for storytelling.

In addition to character background, the books also told a short story about the characters that children could imitate in play afterward.

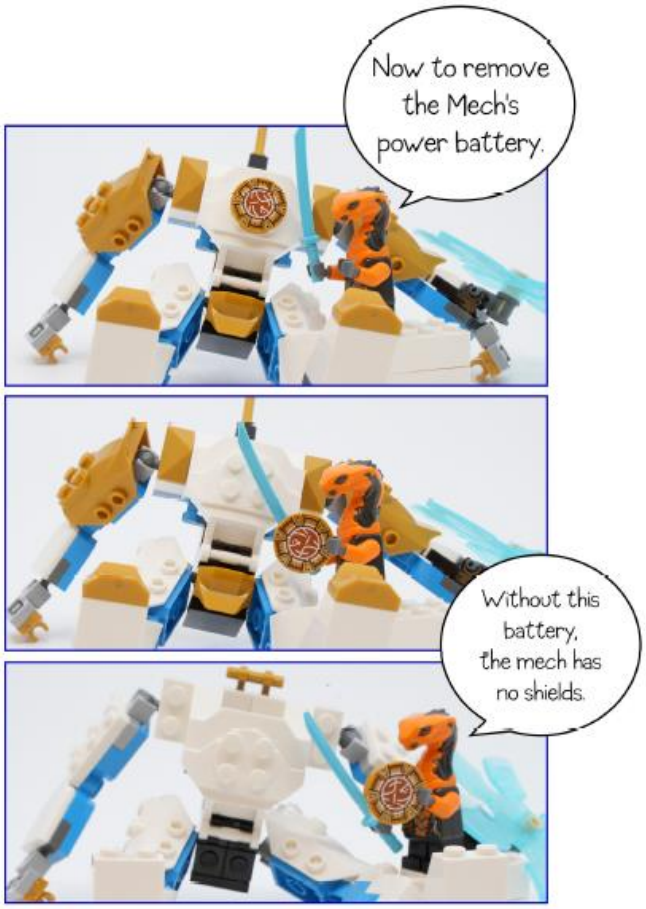

Figure 9. The comic story reveals details about the toy mech.

The final version of these backgrounds was positively received. For example, Emily repeatedly called her brother over to listen to her describe each new character: “Kyle, look, it says she is afraid of crabs.” Noah, upon reading about how to remove the robot’s battery, commented, “This will help me figure out how the battle will go.”

The children who engaged in play repeatedly referenced the newly learned character traits. Cece received a LEGO set that included a turtle named Turbo and a LEGO figure named Emma. She also received a prototype book that described Turbo and Emma in detail. After reading the book, Cece built a home and explained that it was for Turbo, the turtle, and the turtle’s mother. She added a hollow rock full of ice cream, Turbo’s favorite food. In addition, Cece gave the LEGO figurine, Emma, a life jacket in case she got sucked away by the ocean, which was another theme in the book. Then, Cece acted out how Emma used the Turtle Protection Vehicle to scare away the crabs, which happened in the book. Cece also gave Emma a trash can to clean up the ocean, which was an extension of the story – Emma’s original job was to protect turtles.

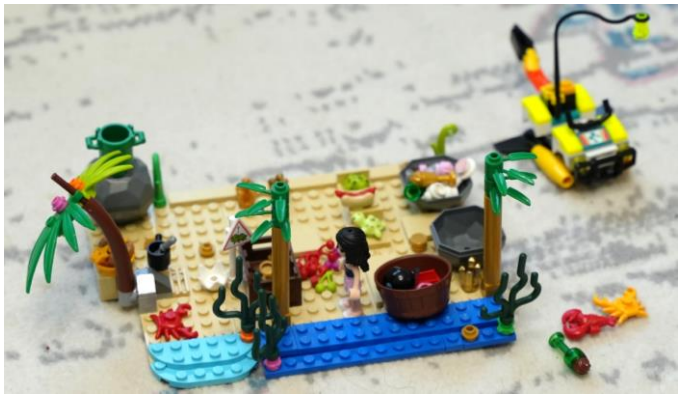

Figure 10. Inspired by the story, Cece builds a whole world for a baby turtle, referencing the story in her choices.

B. Make the story fully imitable with the child’s toys (Iteration)

A key part of play is imitation, where children repeat what they have read or seen previously (N. E. Miller & Dollard, 1941). In advanced forms of imaginary play, children can use props as replacements for key story elements; for example, a banana could act as a phone. This skill of treating objects as symbols of other objects is viewed as a crucial skill in play (Vygotsky & Cole, 1978). However, in our observations, children preferred the story to be a literal replica of their available toys to help inspire play.

In some prototype stories, we added a camping set, soda cans, or dragon that the children did not have included with their toy. Children did not reference these external settings and characters in their play. In one story, we included a jail cell as a setting. In the interviews, the children actively complained that they did not have that jail cell to help tell their story. The young readers explained that without the referenced pieces, they could not repeat the story.

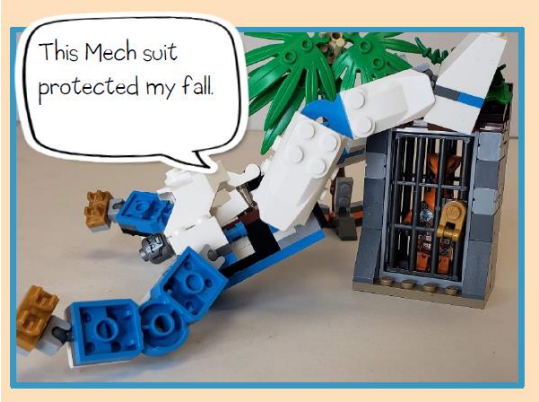

Figure 11. Children were upset that they did not have access to this jail for their own storytelling.

When we asked children if they would prefer exact images of the toy models in the story or more abstract comic book drawings, children consistently responded that they would prefer to see direct pictures of their toys – they wanted images they could imitate.

C. Provide open-ended prompts for storytelling (Joy/Curiosity)

In play tutoring, a teacher provides prompts for children to model play: “Pretend this is a phone and you are calling me” (Smilansky, 1968). In our prototype books, we added open-ended prompts as well.

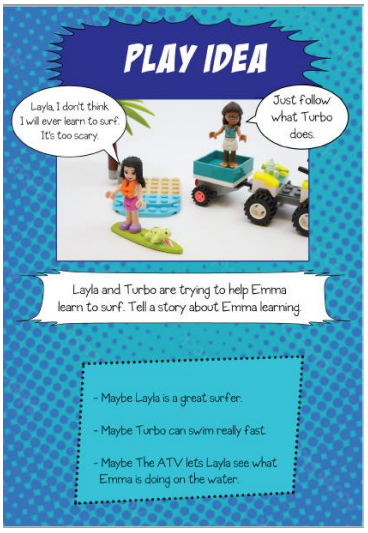

Figure 12. A storytelling prompt given at the end of the book.

Children reported that they liked these prompts, and when they engaged in play, it was often in direct response to the prompts. For example, “What kind of home should Layla and Emma build for Turbo and his mom?” prompted Cece to build the home described previously. “Can you help Zane find a way to overcome his fear and defeat the radical reptiles?” prompted Elijah to put a group of snakes inside a LEGO robot. In response to, “Help tell a story of Zane bringing the chest down Fridge Mountain back to the dojo,” Coulomb put the treasure chest in the robot and had the treasure chest drive itself home.

D. Prompt the children to take digital photos (Active)

Digital cameras can help children self-reflect and retell stories without having to write (Byrnes & Wasik, 2009; Yoon, 2014). In our last week of testing, six children read both a physical and digital book with a camera, which can be accessed here: https://wonder.io/book/902/1.

In the digital book, we included interactive questions that previous research showed could engage readers in a playful manner (Hedman, 2021). Reconfirming our previous findings, our emotional sensors showed the same expected playful engagement with the digital book. Specifically, we found that children emotionally engaged with answering questions during the story, making the reading experience more playful.

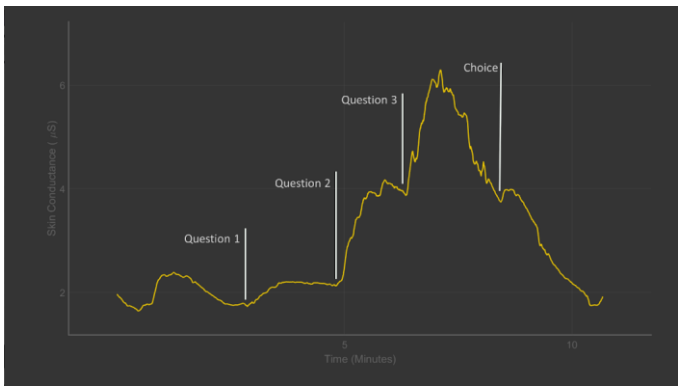

Figure 13. Quentin's skin conductance spikes during reading with interactive questions and prompts.

The last question in our digital book asked the child to “Pretend your phone is a drone and take a picture of the monkeys being silly.” All six children played with the LEGO toy during this prompt. Quentin placed the monkeys in the tree lying down and eating a banana. Nolan pretended the monkeys were taking a selfie. Lucy and Ella took a picture of the monkeys juggling bananas as they made sound effects.

Figure 14. A page in a digital story asks children to act out a story and take a picture.

Upon seeing Emma and Lucy's success with the camera, we asked them to make a movie with the camera. Their impromptu movie lasted for four minutes and included additional details from the books:

Lucy: "What are you doing here?"

"I'm taking pictures of the turtles."

"You shouldn't be here. That's wildlife rules."

When the six children read our physical book, none of them played with their model, but the digital camera engaged all of them to play and storytell with their toys. Integrating a digital camera into a book is a strong way to facilitate the transition from reading to play and encourage children to storytell.

E. Include adults and siblings in play (Social)

A key pillar of learning through play is having a social component. While many of the children we worked with read and played alone, which is often the case when practicing reading, at other times they read and played with an adult or sibling. These peer interactions with siblings, parents, or the researcher were often much more emotionally engaging than the reading itself.

When Niki was storytelling about a LEGO mech stomping a figurine in half (Figure 15), the sensors recorded a small emotional reaction. When his younger brother asked what was happening, Niki reacted with a larger response. An even larger response occurred when the researcher asked about Niki's play.

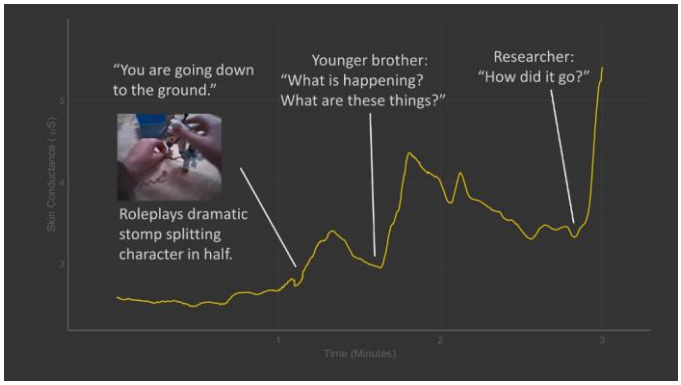

Figure 15. Niki emotionally responds to his brother and researcher during play.

When Evan played with his mother, she pretended the roasted chicken was fighting the mech, and the roasted chicken was going to throw drumsticks at it. One of Evan's largest emotional responses during play was to his mother's provocation (Figure 16).

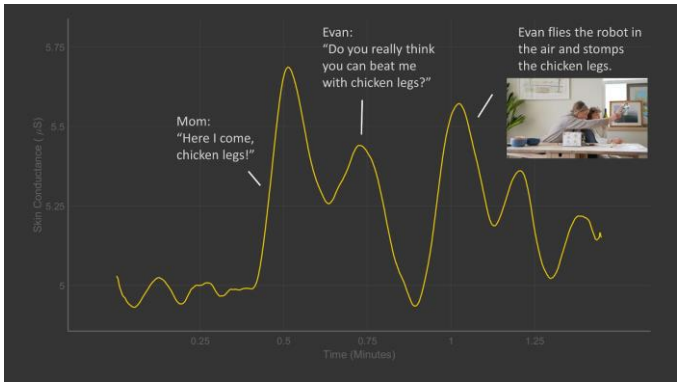

Figure 16. Evan emotionally reacts to his mother's play prompt of attacking chicken legs.

Similarly, Cece only began building Turbo's home when her mother emphasized the prompt in the text of "Do you want to build that home?"

After Gracie and Kaya finished reading, Kaya sat still, staring out into space, while Gracie, the younger sister, played. Gracie made a joke with her toy, which had the effect of inviting Kaya to join in and tell a story about the toy's brand-new outfit (an important part of the story).

Gracie, moving a small bunny LEGO character:

> *"Looks like I've got to go to the bathroom. Pssss. Aaahh!"*

Laughter from Gracie and Kaya.

Gracie: "The bunny peed on the shoe!"

Kaya: "My brand-new outfit, my brand-new shoes, my brand-new everything!"

Figure 17. Kaya and Gracie laugh and play together after reading a prototype book.

Books that encourage reading and playing together with parents or peers are more likely to engage children in play by bringing the social aspect of play to the forefront.

CONCLUSION

Our study on how books can encourage learning through play was explorative – we did not know what we were going to find. We discovered that in our hands-off research setting, children often did not have an innate desire to play. Our five design recommendations created a base for how media can better encourage children to play, but we believe this is just the beginning. Telling a story about monkeys juggling bananas is a start, but we hope to see deeper, more enriching forms of play in the future. How could a book encourage a child to retell the story from a different perspective, or combine the story's themes with the child's own point of view? Reading comprehension is more than just remembering a character, and we believe play can help children practice more advanced literary

skills.

Most of the studies we reviewed looked at how to add learning to play, with an assumption that children will be imaginary playing, storytelling, and creating naturally. Our observations showed that children automatically playing should not be assumed, especially older children. Books are typically designed as passive, non-play activities, but there may be benefits to directly encouraging play within the text. Asking how to design learning to foster play is a new way to reframe the discussion of adding learning to play. How can we encourage play with television, apps, dinnertime, math class, or even recess? Future research should continue to explore how we can empower children to be playful all day, not just during preschool playtime.

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